

AWS re:Invent

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ARC328-R

AWS Well-Architected best practices: Let's talk implementation

Rodney Lester

Customer Technical Program Manager
Amazon Just Walk Out Technology

Mahanth Jayadeva

Resilience Architect
AWS Resilience Hub



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Agenda

Are you Well-Architected?

Well-Architected Framework reviews (WAFR)

Best practice implementation

Q&A

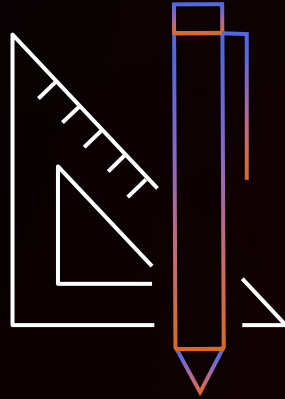
When you look at the
workloads your team is building,
can you answer the question:

“Are you Well-Architected?”

AWS Well-Architected



Learn

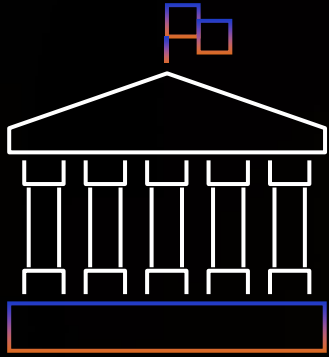


Measure



Improve

AWS Well-Architected Framework



Pillars & lenses



Design principles

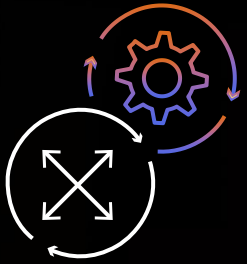


Questions



Best practices

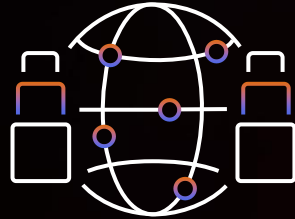
Pillars of AWS Well-Architected



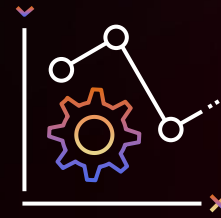
Operational
excellence



Security



Reliability



Performance
efficiency



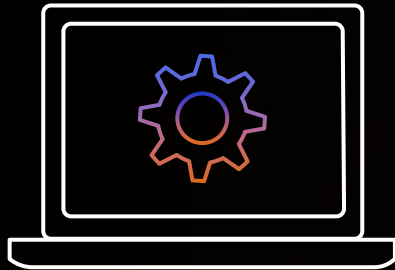
Cost
optimization



Sustainability

Well-Architected Framework review (WAFR)

Identify key workloads



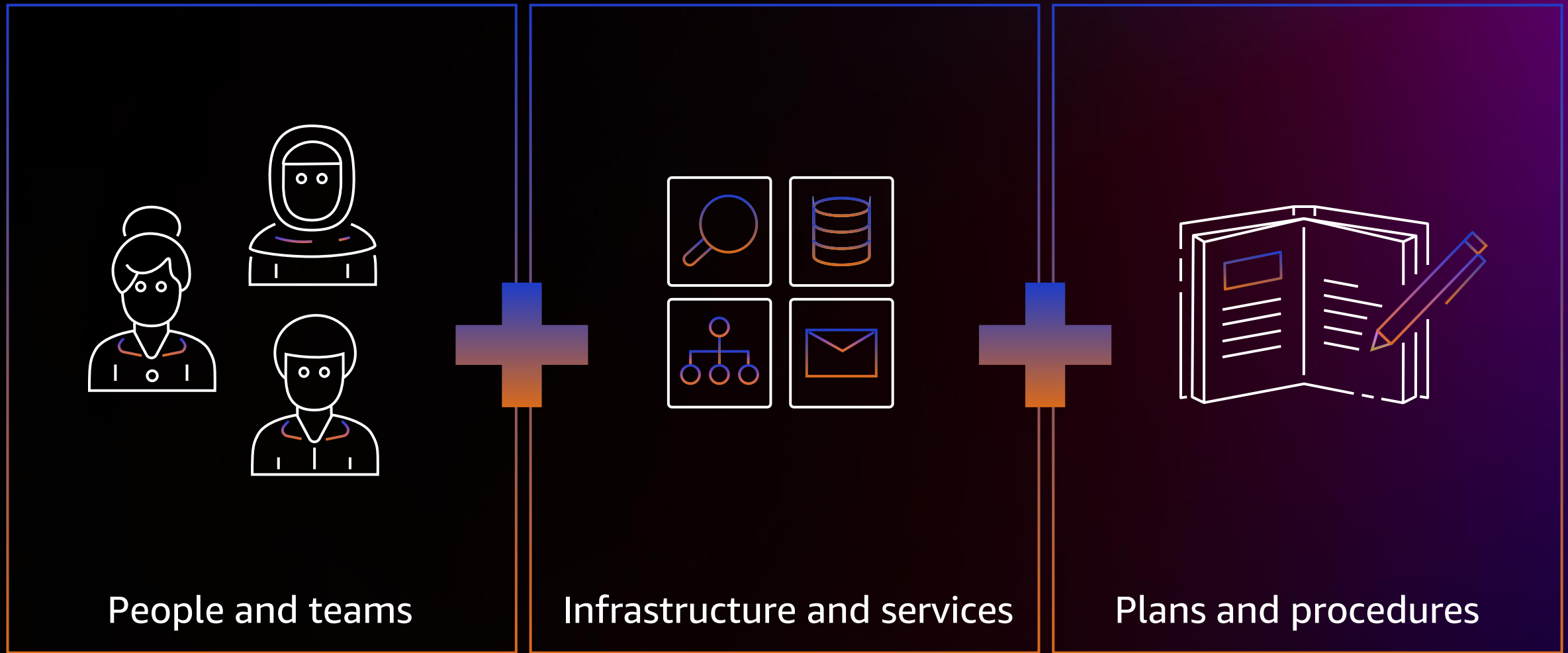
Identify workload sponsor



Review using WA Tool



What is a workload?



Set of components that together deliver business value

AWS Well-Architected Tool

REL 2. How do you plan your network topology? [Info](#)

Ask an expert [↗](#)

Workloads often exist in multiple environments. These include multiple cloud environments (both publicly accessible and private) and possibly your existing data center infrastructure. Plans must include network considerations such as intra- and inter-system connectivity, public IP address management, private IP address management, and domain name resolution.

☐ Question does not apply to this workload [Info](#)

Select from the following

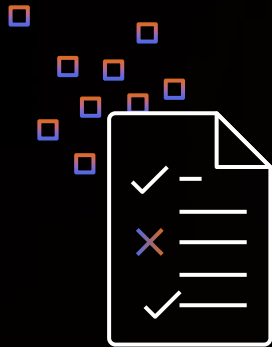
- ☒ Use highly available network connectivity for your workload public endpoints [Info](#)
- ☒ Provision redundant connectivity between private networks in the cloud and on-premises environments [Info](#)
- ☒ Ensure IP subnet allocation accounts for expansion and availability [Info](#)
- ☒ Prefer hub-and-spoke topologies over many-to-many mesh [Info](#)



AWS Well-Architected

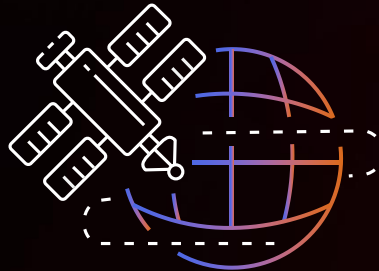
Intent of a review

Not an audit



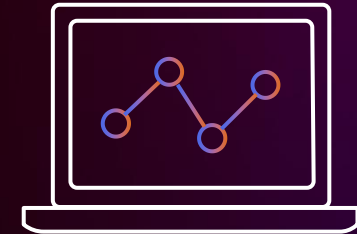
Working together to
improve

Not theoretical



Pragmatic,
proven advice

Not a one-time check



Throughout
lifecycle

Operational excellence

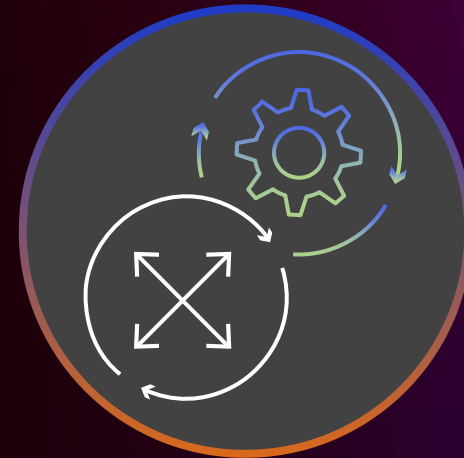
Perform operations as code

Make frequent, small, reversible changes

Refine operations procedures frequently

Anticipate failure

Learn from all operational failures



Security

Implement a strong identity foundation

Enable traceability

Apply security at all layers

Automate security best practices

Protect data in transit and at rest

Keep people away from data

Prepare for security events



Reliability

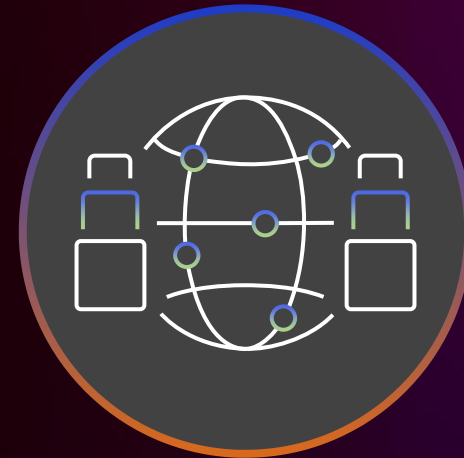
Automatically recover from failure

Test recovery procedures

Scale horizontally to increase aggregate workload availability

Stop guessing capacity

Manage change in automation



Performance efficiency

Democratize advanced technologies

Go global in minutes

Use serverless architectures

Experiment more often

Consider mechanical sympathy



Cost optimization

Practice Cloud Financial Management (CFM)

Adopt a consumption model

Measure overall efficiency

Stop spending money on undifferentiated heavy lifting

Analyze and attribute expenditure



Sustainability

Understand your impact

Establish sustainability goals

Maximize utilization

Anticipate and adopt new, more efficient hardware and software offerings

Use managed services

Reduce the downstream impact of your cloud workloads



Resources



AWS Well-Architected Framework



AWS Well-Architected Tool



AWS Well-Architected Partners



AWS Well-Architected Labs



Thank you!

Rodney Lester

 [linkedin.com/in/rodneylester](https://www.linkedin.com/in/rodneylester)

 @SanDiegoPoke

Mahanth Jayadeva

 [linkedin.com/in/mahanthjayadeva/](https://www.linkedin.com/in/mahanthjayadeva/)

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